THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Life Processes in Plants** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which observation shows that a plant is making more food than it uses? 1

(a) it gains mass and grows **(b)** it turns pale
(c) it stops growing **(d)** it loses mass

2. Bread mould is a: 1

(a) insectivore **(b)** autotroph **(c)** saprotroph **(d)** parasite

3. Which of the following does not increase the rate of transpiration? 1

(a) wind **(b)** high temperature **(c)** dry air **(d)** high humidity

4. A plant in deep shade may not grow since: 1

(a) photosynthesis is less than respiration **(b)** it gets more CO2
(c) roots are weak **(d)** it gets more light

5. The yellow threads on a hedge are: 1

(a) moss **(b)** fungi
(c) lichen **(d)** Cuscuta, a parasite

6. Which of the following will be the first to wilt in hot weather? 1

(a) a plant with many large leaves **(b)** a cactus
(c) a plant with small waxy leaves **(d)** a pine

7. Yellowing of leaves can be due to a shortage of: 1

(a) carbon dioxide **(b)** sand
(c) nitrogen or magnesium **(d)** oxygen

8. A plant that makes its own food and also catches insects is: 1

(a) insectivorous **(b)** lichen **(c)** saprotrophic **(d)** parasitic

9. **Assertion (A):** Transpiration is the loss of water vapour from the leaves through stomata. 1
Reason (R): Transpiration takes place only through the roots of a plant.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Xylem carries the food made in the leaves to all parts of the plant. 1
Reason (R): Phloem carries the food made in the leaves to other parts.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is alcohol used to remove chlorophyll in the starch test? 2

12. Explain the mutual benefit in the lichen association. 2

13. Why are insectivorous plants usually found in marshy soil? 2

14. Why does a plant in sunlight produce more oxygen than it uses? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why are greenhouse owners increasing carbon dioxide levels and using lamps? 3

16. How does Rhizobium help to make plant protein without fertiliser? 3

17. Explain why the green plants are called 'primary producers'. 3

OR

Explain why spinach grown in magnesium-poor soil appears yellowish. 3

18. Explain why plants growing near a window grow towards light and what it shows. 3

OR

How can you show that roots absorb water, using a stained water experiment? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ravi covers a leafy branch of a plant growing in a pot with a transparent polythene bag and keeps it in the sun. After some hours, tiny water droplets appear inside the bag. 5
(i) Name the process that released the water vapour.
(ii) Name the tiny pores through which it was lost.
(iii) Name the tissue that carries water from roots to leaves.
(iv) State one benefit of this process to the plant.

20. Discuss how deforestation and urbanisation affect photosynthesis, the water cycle and carbon dioxide levels, and suggest remedies. 5

OR

Explain how a pitcher plant, a Venus flytrap and a sundew are adapted to catch insects and why this does not make them animals. 5

END OF QUESTION PAPER — **Total: 40 Marks**